

SUMMARY

Solana-focused software & SVM engineer specializing in production-ready on-chain systems and full-stack dApps across DeFi, DePIN, DAOs, infrastructure, and tooling. Architect and deploy Rust/Anchor programs with Solana web3.js integrations, leveraging Pinocchio-style zk systems to drive scalability and privacy innovations. Additional expertise as a software & GenAI engineer (LangChain, RAG, LLM tooling) and systems/network administrator, emphasizing robust security across distributed networks and applications. Solana Turbin3 Builders & Advanced SVM alum; zk-compression hackathon winner; Arcium Fellowship member

SKILLS

- **Programming Languages:** Rust, C++, C JavaScript/TypeScript, Python
- **Blockchain & Web3:** Anchor framework · BPF programs in Rust · PDAs · Cross-Program Invocations (CPI) · Borsh & zero-copy serialization · Program lifecycle management (architecture to deployment) · Vault contracts & staking solutions · AMM implementations · Lending/borrowing primitives · Escrow patterns · Solana CLI · solana-test-validator · devnet/testnet workflows · Unit/integration/fuzz testing (Mollusk, LiteSVM, Surfpool) · Performance optimization & compute cost reduction · Security best practices (invariants/audits) · Solana web3.js · React for dApp UIs · TypeScript integrations · Phantom & Solflare wallet integrations · SPL Token · Metaplex NFTs · Serum DEX basics · Solana Pay · zk-compression (Light Protocol) · Arcium MPC · Transaction-conflict analysis & mitigation (read/write set design) · Jito Bundles & MEV
- **Zero-Knowledge & Cryptography:** zk-SNARKs · Merkle trees/proofs · Poseidon hash
- **AI/ML & Generative AI:** TensorFlow · PyTorch · Hugging Face · scikit-learn · LangChain/LangGraph · RAG · Prompt engineering · Model APIs (OpenAI, LLaMA, Claude, Mistral)
- **Databases & Web Development:**
 - **Databases:** MongoDB · PostgreSQL · Astra DB · Pinecone · Walrus Storage.
 - **Backend:** Node.js (NestJS, Express) · Django · FastAPI
 - **Frontend:** React · Next.js
- **Tools & DevOps:** Docker, Git, GitHub, CI/CD, Azure , Jenkins

EXPERIENCE

Solana Program Developer (Turbin3 Builders & Adv SVM Alum) –Turbine3 – Remote - Apr 2025 – Present

- Architected and deployed Anchor-based Solana programs in Rust, handling PDAs, CPIs, Borsh serialization; developed on-chain escrow, AMM, vault, staking contracts, quadratic voting, and decentralized marketplace for DeFi primitives.
- Minted NFTs/fungible tokens via Metaplex/Umi; leveraged SPL Token standards and zk-compression (Light Protocol) for efficient.
- Wrote TypeScript unit tests using LiteSVM , Mollusk.
- Analyzed Agave banking stage code; implemented new syscalls/sysvars with benchmarking to optimize SVM performance.
- Built optimistic rollup prototype for cost-efficient, reliable DeFi; collaborated on team ZK rollup; contributed PR to Agave repo enhancing banking stage scheduling reliability (under work).

AI/ML Developer (Full Time) – Niloom AI, New York, Remote - June 2024 – Present

- Integrated multiple LLMs (OpenAI, Mistral, LLaMA) using **LangChain** and **LangGraph**.
- Implemented vector embeddings and similarity search techniques for efficient data retrieval.
- Engineered autonomous workflows using chaining & prompt engineering for task orchestration
- Applied **RAG** techniques to optimize LLM responses, reducing average response time.
- Created and ran unit tests (RAGAS) to ensure reliability of AI models and API integrations.

MERN Stack Developer (Internship) – Ark X - January 2024 – June 2024

- Developed full-stack MERN applications (MongoDB, Express.js, React, Node.js).
- Engineered high-performance server-side applications with real-time data processing.

Software Engineer (Bootcamp) – Alx Africa February 2023 – April 2024

- Engineered full-stack web applications with React/Next.js, Node.js/Express.js
- Built and deployed server-side applications with Python frameworks (**Django, Flask**).
- Designed and maintained SQL and NoSQL databases, optimizing performance.

Junior Pentester (Internship) - April 2023 – July 2023

- Conducted penetration testing and vulnerability scans; built IPS/IDS for proactive threat detection; and implemented remediations, reducing vulnerabilities.

EDUCATION

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| • Turbin3 builders cohort and Adv Svm | Turbin3 | Apr 2025 – aug 2025 |
| • Fullstack MERN Program | JobInTech | Jan 2024 – Jun 2024 |
| • Software Engineering Program | ALX Africa · | Feb 2023 – Apr 2024 |
| • Bachelor in Engineering and Security of Networks | Higher School of Technology, | Aug 2022 – July 2023 |

PROJECTS

- **RLock (DeFi SDK):** Parallel Solana execution toolkit bundling complex intents into conflict-aware, optimized transactions for 95%+ success rates and 90%+ fee savings—leverages OCC DAG execution, Agave banking stage enhancements (syscalls/sysvars/PR for reliable scheduling); built optimistic/ZK rollup prototypes; integrates with @solana/web3.js for seamless DeFi flows.
- **ChainSensors (DEPIN):** ChainSensors: Solana-based decentralized IoT marketplace enabling sensor owners to monetize air-quality/temp/logistics data—on-chain Anchor/Rust smart contracts w/ PDAs/zk-compression; off-chain Walrus storage/TLS secure transport; full-stack Next.js/NestJS/Anchor/web3.js/Phantom integration.